Assignment 4.1 Exercise

In this assignment, you will extend the concepts demonstrated in the lab portion to analyze income levels within a specific ethnic group: Native Americans. The goal is to understand how applying differential privacy impacts our analysis when focusing on a smaller, often underrepresented population group.

Specifically, you will explore how the Laplace mechanism affects the **income level counts** for Native Americans. Education and other government funding often depend on accurate data about income levels, and differential privacy can introduce noise that may distort these figures. Your task will be to determine how varying the privacy budget (epsilon) changes the accuracy of these counts and to reflect on the broader implications of these changes, particularly when allocating resources to underrepresented groups.

Step 1: Calculate Original Counts for Income Levels

Display the original income counts for Native Americans:

Low: 45
Middle: 39
High: 27

Step 2: Apply Laplace Mechanism to Add Noise

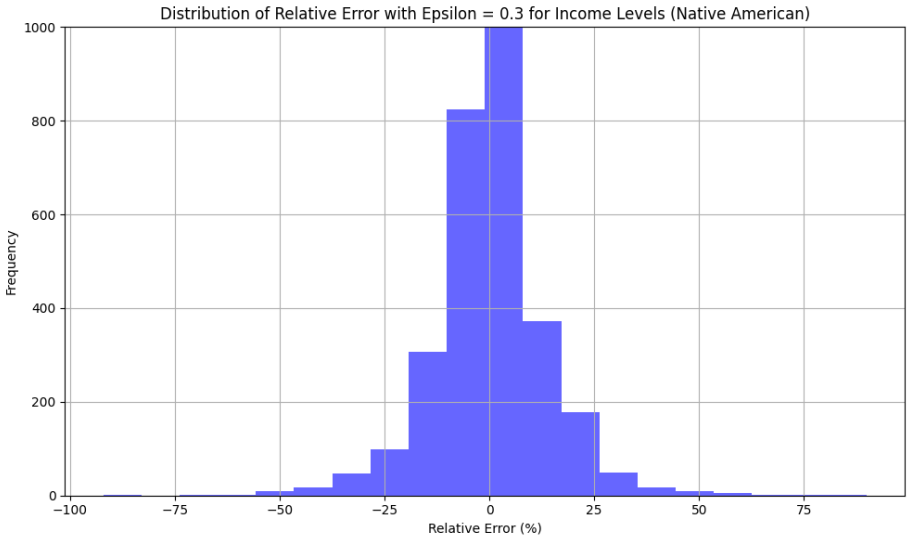
Apply the Laplace mechanism to add noise with at least two epsilon values.

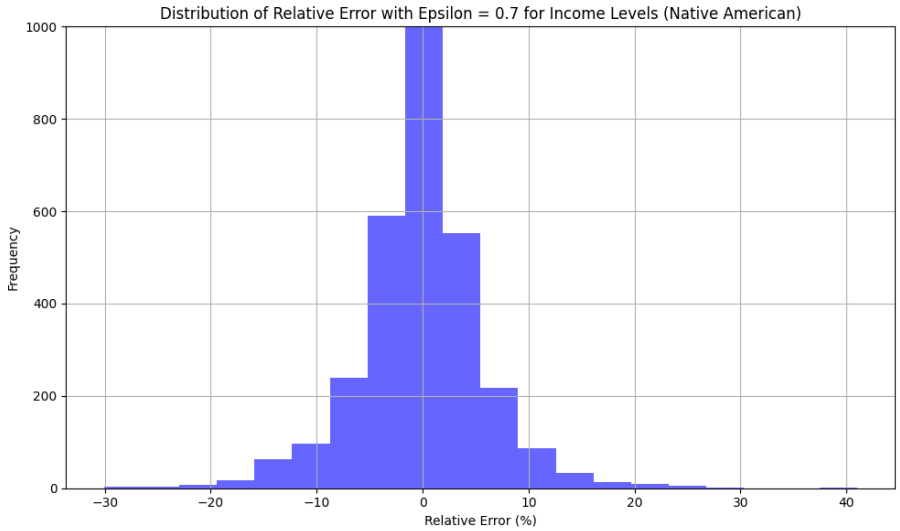
Attach a screenshot or output of your code here:

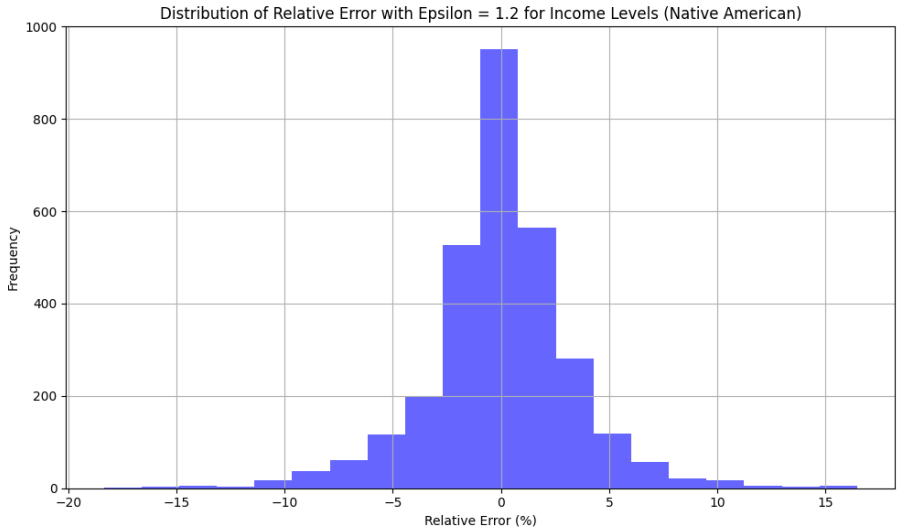
```
Change in Representation with Epsilon = 0.3:
Income Level: Low
 - Original Count: 45
 - Noisy Count: 46.59138837627687
 - Change in Representation (%): 3.54%

Income Level: Middle
 - Original Count: 39
 - Noisy Count: 37.85785640692149
 - Change in Representation (%): -2.93%

Income Level: High
 - Original Count: 27
 - Noisy Count: 34.671424588341196
 - Change in Representation (%): 28.41%


Change in Representation with Epsilon = 0.7:
Income Level: Low
 - Original Count: 45
 - Noisy Count: 45.48138662071411
 - Change in Representation (%): 1.07%

Income Level: Middle
 - Original Count: 39
 - Noisy Count: 43.57230733631852
 - Change in Representation (%): 11.72%

Income Level: High
 - Original Count: 27
 - Noisy Count: 25.633705732951732
 - Change in Representation (%): -5.06%


Change in Representation with Epsilon = 1.2:
Income Level: Low
 - Original Count: 45
 - Noisy Count: 45.103259321452
 - Change in Representation (%): 0.23%

Income Level: Middle
 - Original Count: 39
 - Noisy Count: 38.87846750530829
 - Change in Representation (%): -0.31%

Income Level: High
 - Original Count: 27
 - Noisy Count: 27.02843262840094
 - Change in Representation (%): 0.11%
```

Step 3: Generate Relative Error Visualization

Screenshot the generated histograms for relative errors using the Laplace mechanism and paste your output for both below:







Step 4: Observation and Reflection

1. Briefly describe your findings. How does changing the value of epsilon affect the relative error?

Your response:

The results demonstrate how the choice of epsilon affects the accuracy of differentially private data. At ε = 0.3, strong privacy introduces significant noise, leading to large distortions, particularly in the high-income group, which increased by 28.41%. At ε = 0.7, the distortion is reduced but still noticeable, especially in the middle-income group. At ε = 1.2, the noisy counts are nearly identical to the original values, indicating high accuracy. These findings show that smaller population groups are more sensitive to noise, as even small absolute changes result in large percentage differences. Therefore, while lower epsilon values provide stronger privacy, they can reduce data utility and potentially lead to misleading conclusions.

2. Is there an epsilon value that produces useful results while still providing privacy protection?

Explain your reasoning.

Your response:

Yes, there are epsilon values that strike a balance between privacy and usefulness. In this analysis, values around 1 produce useful results by keeping the relative error small while still adding noise to protect individual privacy. Lower epsilon values introduce excessive noise, making the data less reliable, while higher values improve accuracy but reduce privacy protection. Because this dataset has a relatively small sample size, a slightly higher epsilon can improve the usefulness of the results. For this reason, an epsilon value of 1.2 provides a better balance than 1 for this particular dataset.

3. Discuss how the trade-offs between privacy and data utility in differential privacy might impact smaller communities in a dataset (i.e. deflated or over-inflated representation). How might the added noise impact decision-making, especially in the context of census data?

Your response:

The trade-off between privacy and data utility in differential privacy can have a significant impact on smaller communities within a dataset. Because these groups have smaller counts, the noise added by mechanisms like the Laplace mechanism can lead to large relative errors, resulting in either underrepresentation or overrepresentation. For example, a small increase or decrease in count can translate into a large percentage change, which may distort the true characteristics of the community. In the context of census data, this can affect decision making by misrepresenting income levels, population size, or resource needs. As a result, funding for programs such as education, healthcare, and housing may be incorrectly allocated, either depriving communities of needed resources or allocating resources inefficiently. Therefore, while differential privacy protects individual data, it is important to carefully choose privacy parameters to avoid disproportionately impacting smaller or underrepresented populations.